DARPA Lift Challenge | Flight-control support

Independent remote support for a competing heavy-lift UAV team.

DARPA Lift Challenge | UAV Flight-Control Engineering

Independent technical support for a competing U.S. heavy-lift UAV team



TECHNICAL PORTFOLIO

ArduPilot flight-log analysis, PID and attitude control, vibration, EKF, ESC telemetry and field testing

2026 DARPA Lift Challenge • United States

Remote engineering advisory • Multi-aircraft flight-test campaign

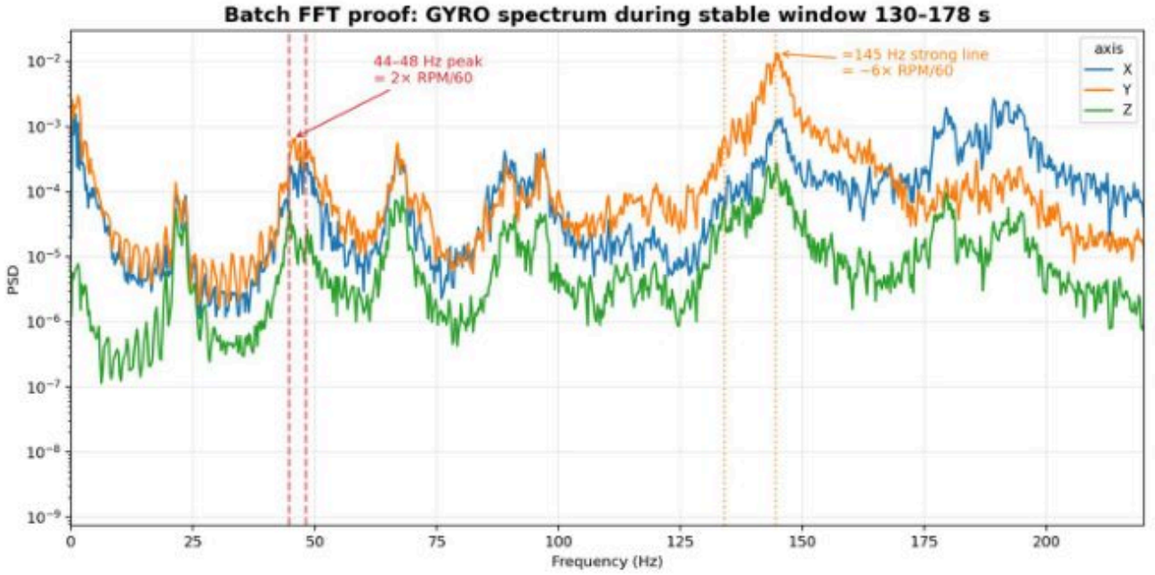
Eng Achraf RAHIBE

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00000031.bin | Filter and FFT investigation

The original gyro spectrum drove targeted notch recommendations before changing rate gains



Filter targets identified

RPM-driven notch
22 Hz start; covers 22–24 Hz + 2× harmonic

Second narrow notch
≈145 Hz spectral peak, from FFT evidence

Validation sequence
Mechanical checks → short log → rate PID

Recommended values from the original engineering report, not claimed as flight-verified settings.

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Engineering deliverables & documented results

What was analysed, documented and recommended across the flight-test campaign

01 FLIGHT-LOG FORENSICS

- DataFlash ATT / RATE / PID / EKF / IMU / VIBE / BAT correlation
- Crash chronology: fault indicators, attitude departure, then mixer saturation
- Separated ground-run warnings from valid airborne tuning evidence

02 PROPULSION & TELEMETRY

- Hobbywing X11 G2 / CUAV X25-EVO CAN, NodeID and ThrottleID review
- Measured four ESC RPM streams in Log 31; 22.3–24.1 Hz hover fundamental
- Flagged ~32% current difference between motor pairs in stable hover

03 FILTERS, STABILITY & PID

- Log 13: 14 kg / 70 kg cases with 1,549 / 3,462 clipping events
- Notch recommendations grounded in FFT peaks and batch-IMU data
- QuickTune Log 49: saved Roll D 0.00250 → 0.00344; documented aborts

04 CLIENT SUPPORT & HANDOFF

- Editable engineering slide decks, diagnostic plots and parameter reviews
- Multi-hour flight-test calls, symptom interpretation and decision support
- Stepwise next-flight checklists and validation plans instead of speculative tuning

Outcome: documented faults, measured subsystem behavior, and practical next-test decisions—not unverified claims of final flight qualification.